



```
sketch_dec13c
lcd.print("BT CONTROLLED");
lcd.setCursor(0,1);
lcd.print(" ROBOT");
delay(1000);
pinMode(pin, INPUT);
pinMode(pin1, INPUT);
pinMode(led1, OUTPUT);
pinMode(led2, OUTPUT);
pinMode(led3, OUTPUT);
pinMode(led4, OUTPUT);
pinMode(led5, OUTPUT);
pinMode(led6, OUTPUT);
pinMode(led7, OUTPUT);
pinMode(led8, OUTPUT);
pinMode(led9, OUTPUT);
pinMode(led10, OUTPUT);
digitalWrite(led1,LOW);
digitalWrite(led2,LOW);
digitalWrite(led3,LOW);
digitalWrite(led4,LOW);
digitalWrite(led5,LOW);
digitalWrite(led6,LOW); digitalWrite(led7,LOW); digitalWrite(led8,LOW);

digitalWrite(led9,LOW); digitalWrite(led10,LOW);
}
void loop()
{
  //Serial.println("BT CONTROLLED");
}
```

Fig. 4: Code snippet one



Fig. 5: LCD fixed in the robot

Chemi-resistor as the detection is based upon change of resistance of the sensing material when the gas comes in contact with the material.

Using a simple voltage divider network, concentrations of gas can be detected. MQ2 works on 5V DC and draws around 800mW. It can detect LPG, smoke, alcohol, propane, hydrogen, methane, and car-

bon monoxide concentrations anywhere from 200 to 10000ppm.

Humidity sensor. A humidity sensor is an electronic device that measures the humidity in its environment and converts its findings into a corresponding electrical signal. A humidity sensor is used to measure and report moisture. Humidity sensors vary widely in size and functionality.

Relay. This electro-mechanical device is used to make or break an electrical connection. It has a flexible

moving part that is controlled electronically through an electromagnet. Basically, it is just like a mechanical switch that is controlled with an electronic signal instead of manually turning it on or off.

LCD. A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals combined with a polarizer. Liquid crystals do not emit light directly; they use a backlight or reflector to produce images in colour or monochrome.

Motor. A direct current (DC) motor converts electrical energy into mechanical energy. DC motors take electrical power from direct current and convert it into mechanical rotation.

MOSFETs. The P55NF06 metal oxide semiconductor field effect transistor (MOSFET) is a 3-terminal n-channel silicon device with current conduction capability of about 50A, fast switching speed, low on-state resistance, breakdown voltage rating of 60V, and max threshold

voltage of 4V. Thin semi 50N06 is 3-terminal silicon device with current conduction capability of about 50A, fast switching speed, low on-state resistance, breakdown voltage rating of 60V, and max threshold voltage of 4V. It is mainly used for electronic ballasts and low-power switching mode power appliances.

Pump. The motor pump is used to move the liquids/gases from one place to another through its mechanical action. Different types of mechanical pumps are available in the market to serve in an extensive range of applications like water pumping from wells, filtering of an aquarium's or pond's water, and in industries like automobile, energy, and medical.

Power supply. The Arduino Uno can be supplied power with the help of a USB cable or an external power supply. The external power supplies include AC to DC adaptor and a battery. The adaptor can be connected to the Arduino Uno by plugging it into the power jack of the Arduino board. Similarly, the battery leads can be connected to the Vin and GND pins of the power connector. The suggested voltage range is 7 to 12V.

To prepare the code for the robot, open Arduino and declare variables and pin modes when the sketch starts. Each will power up by importing their libraries. Code's first snippet is shown in Fig. 4.

Construction and testing

An overflow will occur for a millisecond so that the Arduino board begins running the current program. After setting the functions in the code MaincodeCoalrobot.ino upload the code by selecting the right port and board to Arduino. Then connect the components according to the circuit diagram. Follow the steps mentioned below.

Step 1. Upload the source code MaincodeCoalrobot.ino into Arduino Uno board